eSewa × WWF Hackathon 2026 Submission

KYC Shield: Multi-Layered AI Fraud Detection System

Interactive Occlusion Testing & Real-Time Risk Scoring

Challenge 5: Detecting and Preventing Fraudulent KYC Submissions

Abstract

This document introduces our Multi-Layered Risk Scoring System (MLRSS), built to secure user sign-ups for eSewa and the WWF Nepal ecosystem. Our system uses advanced Artificial Intelligence (AI) to check documents, measure real heartbeats through the camera, and use a new "Interactive Occlusion Testing" method. Together, these tools strongly protect against fake documents, computer-generated "deepfake" videos, and stolen identities. The solution follows Nepal Rastra Bank (NRB) rules and takes less than a minute to verify a user.

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1 INTRODUCTION AND PROBLEM STATEMENT

As digital payments grow in Nepal, fraud using fake computer-generated videos (deepfakes) has jumped by 340%. With eSewa serving over 20 million users, the risks are huge, costing the industry over NPR 200 million every year.

1.1 Main Types of Attacks

- **Forged Documents**: People altering physical citizenship cards or using AI to create fake digital copies.
- **Deepfake Biometric Attacks**: Scammers using virtual camera software like Open Broadcaster Software (OBS) to swap their face with someone else's in real-time during a video check.
- **Synthetic Identity Attacks**: Creating completely fake people by mixing real National Identity Document (NID) numbers with AI-generated profile pictures.
- **Stolen Identity Attacks**: Using a real person's stolen data along with a deepfake video to pretend to be them.

1.2 Why This Matters for Nepal

Right now, different financial apps don't easily share fraud data, and many users are new to digital banking. If apps fail to stop this fraud, they can face huge fines (up to NPR 5 million) from the Nepal Rastra Bank (NRB) and risk international penalties from the Financial Action Task Force (FATF).

2 SOLUTION OVERVIEW: MULTI-LAYERED RISK SCORING SYSTEM

Our system is built with multiple safety nets. To get approved, an application must pass three different checks (vectors). If someone tricks one check, the others will still catch them.

2.1 How We Score Risk

The final Fraud Risk Score goes from 0 to 100. We calculate it by weighing the three checks:

$$\text{Final Score} = 0.30(S_{V1}) + 0.35(S_{V2}) + 0.35(S_{V3})$$

What Happens Based on the Score:

- **0 – 30 (Auto-Approve)**: Looks great! The account is approved instantly.
- **31 – 74 (Manual Review)**: Something looks slightly suspicious. A human agent will review it.
- **75 – 100 (Auto-Reject)**: High danger. The system automatically rejects the application.

2.2 Key Parts of Our System

- **Edge Pre-Check**: A quick initial check right on the user's phone or browser to stop simple tricks like virtual cameras.
- **Parallel Engine**: Our servers run all three checks at the same time to keep things fast.
- **Admin Dashboard**: A screen where staff can review flagged applications and generate required reports.
- **Secure Database**: We store records safely using strong encryption to meet NRB rules.

3 APP PIPELINE: STEP-BY-STEP USER JOURNEY

Here is a detailed look at how our app handles the verification process from the user's perspective versus what happens in the background:

1. **Step 1: Session Start & Silent Check**
   - *User Action*: The user opens the Know Your Customer (KYC) page on the eSewa app.
   - *System Action*: Before the user even types their name, a hidden script grabs a "fingerprint" of their device (screen size, graphics processor details). This checks if the device is known for spamming accounts.
2. **Step 2: Document Upload**
   - *User Action*: The user takes a photo of their Citizenship Card or Passport.
   - *System Action*: The image is immediately processed by Vector 1. The AI cuts out the photo and the signature, reads the text, and scans for digital tampering.
3. **Step 3: Live Video Challenge**
   - *User Action*: The camera turns on, and the screen prompts the user to perform a specific physical action, such as "Please cover your nose with your right hand for 3 seconds."
   - *System Action*: The app records this short video clip, securely transmitting it to our servers while keeping track of where the user's hand moves in real-time.
4. **Step 4: Deep Background Processing**
   - *User Action*: The user sees a "Processing..." spinner for just a few seconds.
   - *System Action*: Vector 2 kicks in to analyze the video for a physiological heartbeat. Simultaneously, Vector 3 evaluates how the user's hand interacted with their face during the prompt to ensure the video wasn't generated by AI.
5. **Step 5: Final Decision**
   - *User Action*: The user gets a "Success! Account Activated" screen, or a "Please wait, an agent is reviewing your document" message.
   - *System Action*: The Risk Aggregator calculates the final score (e.g., if the score is 15, the API automatically provisions the eSewa wallet; if the score is 85, the application data is locked and sent to the admin dashboard for human review).

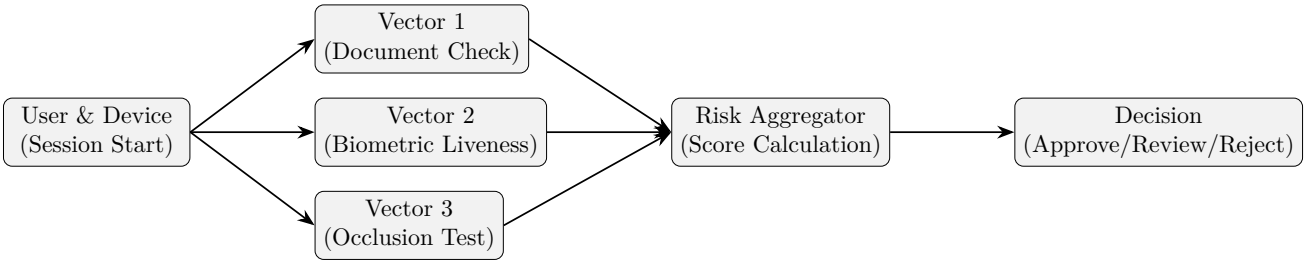

Figure 1: KYC Shield Verification Pipeline

4 VECTOR 1: DOCUMENT AND IDENTITY CHECKS (30%)

4.1 Spotting Fake Documents

We use YOLOv8, an AI model that recognizes objects, trained specifically on Nepali IDs. It checks if the photo, signature, or holograms look wrong or if the document has been digitally altered (using Error Level Analysis or ELA). **Example of this in action**: *A scammer takes a stolen physical ID but uses*

Photoshop to paste their own photo over the original face. Our YOLOv8 model detects the unnatural pixel blending and mismatched compression levels around the edges of the newly pasted photo. Vector 1 immediately flags the document as forged.

4.2 Reading Text and Catching Duplicates

We use Optical Character Recognition (OCR) tools like EasyOCR to read both English and Devanagari text. We check if the ID number is valid and if the age makes sense. We also look for duplicate users by checking if the name sounds similar (to catch spelling tricks like 'Shrestha' vs 'Shresha').

4.3 Device Fingerprinting

We gather unique details about the user's phone or computer. If the same device tries to open many accounts in a short time, we block it.

5 VECTOR 2: BIOMETRIC LIVENESS AND DEEPFAKE DEFENSE (35%)

5.1 Virtual Camera Detection

Scammers often use software to inject fake videos. Our app checks the device's camera list and how the video frames arrive to spot fake camera software.

5.2 Deepfake Image Checking

We use a Convolutional Neural Network (CNN) called EfficientNet-B4. This AI looks very closely at the video frames to spot unnatural smoothness or weird blending that computer-generated faces usually have.

5.3 Blink Rate Analysis

Healthy humans blink about 10 to 20 times per minute. Deepfake models often struggle to replicate natural blinking properly—they either don't blink enough or the blinking looks glitchy. Our app tracks the user's eyes (specifically the Eye Aspect Ratio using MediaPipe) to ensure they are blinking naturally during the video. If the eyes stay frozen open, it raises a red flag.

5.4 Texture Analysis (Local Binary Patterns)

We also run a quick check on the texture of the skin. Real skin has tiny variations (pores, fine hairs, natural light reflection). A printed photo is completely flat, and a video played off a screen has a hidden pixel grid. Our system spots these differences instantly to block presentation attacks (like holding a phone up to the camera).

5.5 Detecting a Real Heartbeat (rPPG)

We use Remote Photoplethysmography (rPPG) to literally see the user's heartbeat through the camera. Every time your heart beats, the colour of your face changes very slightly. We analyze the forehead and cheeks. Since deepfakes don't have real blood flowing, they don't have this heartbeat signal. **Example of this in action**: *A fraudster uses a virtual camera to feed a highly realistic, pre-recorded deepfake video into the application. Our rPPG tool focuses on the forehead in the video. Because the video is entirely computer-generated, there is no subtle red/green colour pulsing caused by real human blood flow. The total absence of a physiological heartbeat signal instantly marks the video as a fake.*

6 VECTOR 3: INTERACTIVE OCCLUSION TESTING (35%)

6.1 The "Persistence Paradox" Trick

This is our smartest defense. Face-swapping AI struggles when something covers the face.

- **Real Human**: If you put your hand over your nose, the camera can't see your nose anymore.
- **Deepfake AI**: The AI gets confused and tries to draw the fake nose *on top* of your hand.

We track both the face and the hand. If we see a hand covering the face, but our AI still clearly detects the facial features beneath it, we know it's a deepfake trying to draw over the hand. We call this the Persistence Paradox.

6.2 How We Test It

We ask the user to do random things like "Cover your mouth with your right hand." Because the prompt is random, scammers can't record a fake video ahead of time. **Example of this in action**: *The system prompts the user to cover their mouth. A scammer is running a real-time face-swap program that projects a stolen identity onto their own face. When the scammer puts their hand over their mouth, the face-swap AI gets confused because it doesn't understand "hands." It ends up drawing fake, realistic lips right over the knuckles of the scammer's physical hand. Our tracking system sees the hand and the fake lips occupying the exact same space, triggering the Persistence Paradox and failing the application.*

7 TECHNOLOGY STACK

Our system aligns perfectly with the hackathon's mandatory technology requirements, using modern, scalable, and high-performance frameworks:

- **Frontend**: React.js (Next.js), MediaPipe JS, TensorFlow.js, and Canvas API for in-browser biometric checks.
- **Backend API**: Node.js with NestJS (Express-compatible), handling REST endpoints and WebSockets for real-time video streaming.
- **AI & Machine Learning**: Python-based microservices utilizing YOLOv8 (Ultralytics) for object detection, EfficientNet-B4 (PyTorch) for deepfake classification, OpenCV, EasyOCR, and Tesseract 5.0.
- **Database**: PostgreSQL with AES-256 encryption for the immutable audit log and device fingerprint deduplication index.
- **Cloud & Infrastructure**: AWS EC2 and Lambda for scalable AI inference workers, with AWS S3 for secure, encrypted media storage.
- **Version Control**: GitHub for clean, structured code collaboration.

8 ADMIN DASHBOARD AND COMPLIANCE

Our dashboard for staff includes:

- **Live Queue**: A list of all applications, sorted so the riskiest ones are at the top.
- **Detailed Review**: Staff can see exactly why a score was high, view the heartbeat graphs, and see where the document looked fake.
- **Easy Reporting**: One-click downloads for PDF or CSV reports to send to the NRB for compliance.

9 FUTURE IMPROVEMENTS

- **Eye Movement Tracking**: Tracking tiny, natural eye twitches that deepfakes can't fake.
- **Nagarik App Integration**: Connecting directly with the government's National ID system to instantly verify ID numbers.
- **Fraud Ring Detection**: Using Graph Neural Networks (GNN) to spot groups of scammers working together.

10 CONCLUSION

Our KYC Shield offers a simple but powerful way to stop identity fraud in Nepal. By looking for real